

Business Problem Statement

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. The management team has noticed changes in purchasing patterns across demographics, product categories, and purchase channels. They are particularly interested in uncovering which factors, such as discounts, subscriptions, promo codes, seasons, or payment preferences, drive consumer decisions and repeat purchases.

You are tasked with analyzing the company's shopping trends dataset (3,900 customer records spanning attributes such as age, gender, item purchased, category, purchase amount, location, size, color, season, review rating, subscription status, payment method, shipping type, discount applied, promo code usage, previous purchases, preferred payment method, and frequency of purchases) to answer the following overarching business question:

"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

Deliverables

- 1 **Data Preparation & Modeling (Python):** Clean and transform the raw shopping trends dataset for analysis.
- 2 **Data Analysis (SQL):** Organize the data into a structured format, simulate business transactions, and run queries to extract insights on customer segments, loyalty, and purchase drivers.
- 3 **Visualization & Insights (Power BI):** Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make data-driven decisions.
- 4 **Report and Presentation:** Write a clear project report summarizing your key findings and business recommendations. Prepare a presentation that visually communicates insights and actionable recommendations to stakeholders.
- 5 **GitHub Repository:** Include all Python scripts, SQL queries, and dashboard files in a well-structured repository.